Supporting information

AI as an Amplifier in Chemistry Laboratories: A SERG-Based Framework for Embodied and Assessable Instruction

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Appendix S4 (SI). Rubrics, Observation Forms, and Interview Protocols

Note. All instruments are designed around the three core challenges—**misalignment**, **instability**, and **missing evaluation**—and aligned with **SERG** indicators (**S**tability, **E**vidence, **R**eusability, **G**rowth). Scores may be linearly rescaled as needed.

S4.1. Rubrics

S4.1.1 Composite Rubric for Student Lab Reports (0–100, six weighted dimensions)

Dimension	4 Excellent	3 Good	2 Satisfactory	1 Needs Improvement	Weight (%)	SERG
Research purpose & problem statement	Problem is clear, testable, and authentically contextualized	Problem mostly clear with some context	Problem basically specified	Problem vague or misaligned with the task	10	E/G
Methods & procedures	Sound design, controlled variables, complete records	Generally sound with minor gaps	Key steps present	Missing key steps or incomplete records	15	E
Data quality	Complete data, good repeatability, outliers identified/handled	Largely complete, acceptable repeatability	Some gaps or mediocre repeatability	Major missing data or unusable	20	E
Uncertainty & error	Sources, estimation, and propagation are clear; numbers reasonable	Sources stated with quantitative estimates	Only qualitative mention of sources	Not addressed or clearly inappropriate	20	E
Results & conclusions	Conclusions consistent with evidence; comparison/discussion sufficient	Mostly consistent with evidence	Somewhat over-assertive	Inconsistent with data	20	E
Reflection & improvement	Evidence-based, feasible improvements proposed	Improvement ideas present	Reflection general	No reflection	15	E/G

Scoring guidance: Two raters score independently and average scores; record disagreements and adjudication rationale for audit (target $\kappa \geq 0.75$).

S4.1.2 "Uncertainty Expression" Micro-Rubric (0–4, for embedded formative feedback)

- **4**: Clearly lists sources (instrument/operation/model), estimation method (formula/propagation), numerical values with units, and discusses impact on conclusions.
- **3**: Lists sources with quantitative estimates; propagation or impact analysis incomplete.
- **2**: Only qualitative mention of sources; lacks quantitative estimates.

- **1**: Not addressed or clearly incorrect.

S4.1.3 Rubric for Public Project Output (Poster/Oral Defense) (0–100)

Dimension	Description	Weight (%)	SERG
Authenticity & driving question	Problem aligns with context; audience is clear	15	G
Evidence-chain completeness	Data → analysis → conclusion are traceable; citations proper	25	E
Visualization & communication	Graphs conform to standards; narrative is clear; time managed	20	E
Collaboration & role allocation	Roles are appropriate; collaboration records are adequate	10	G
Reflection & iteration	Responds to peer-review comments with documented revisions	15	E/G
Ethics & compliance	Privacy de-identification; acknowledgments and sources clear	15	S/G

S4.1.4 Teacher Lesson–Enactment Alignment Rubric (observer; 0–100)

20-pt Dimension	Key checkpoints	SERG
Goal alignment (TPACK)	Technology–discipline–pedagogy coherent; learning outcomes observable	E/R
Activity flow (SAMR)	Layered progression explicit; tasks implementable	R
Assessment enactment (formative)	Self/peer/teacher evidence-chain present	E
Safety & compliance (RAMP)	Tiered guidance for hazardous steps; emergency readiness; de-identification	S
UDL scaffolds	Multiple means of representation/engagement/expression; dual-channel fallback	S/R

S4.2. Observation Forms

S4.2.1 Classroom Process Observation Form (with stability and engagement)

Time Window	Teaching Activity	Student Activity	Evidence Collected	Event Code	Notes
00:00–05:00	Intro/objectives	Respond/ask	Exit ticket screenshot	**P** (Participation)	
05:00–20:00	Data collection/import	Role allocation/record	Raw data sample	**D** (Data)	
20:00–35:00	Analysis/plotting	Discuss/compare	Charts/parameter table	**A** (Analysis)	
35:00–45:00	Presentation/feedback	Peer review/iteration	Peer forms/key talking points	**F** (Feedback)	

Stability log (side panel on same form)

- Failure type: Network / Software / Hardware / Materials (multiple choices)
- Impact duration (min): ___; Was fallback completed within 5 minutes? Yes/No
- Fallback path: Notebook → Excel / Cloud → Local / Pre-generated figures
- Responsibility & disposition: ___

Engagement scale (0–4): 0 none; 1 sporadic; 2 intermittent; 3 sustained; 4 high intensity (incl. questioning, peer support, quality of artifacts)

S4.2.2 Implementation Fidelity Checklist (√ / ×)

- Goal–task–assessment alignment (TPACK): □ Aligned □ Partial □ Not aligned
- SAMR tier declared: □ S □ A □ M □ R (check)
- Formative tools present: □ Self □ Peer □ Teacher □ Rubric
- Dual-channel fallback: □ Local/Cloud □ Script/Excel □ Pre-generated figures
- RAMP: □ Hazard ID □ Risk assessment □ Risk reduction □ Emergency prep
- Privacy compliance: □ Consent & notice □ Data de-identification □ Public release review

S4.2.3 Safety & Compliance (RAMP) Quick Audit

Item	Yes/No	Notes
Hazard-identification checklist posted and briefed		
Tiered guidance for high-risk operations (dual verification)		
Proper storage and labeling of chemicals		
Emergency supplies and procedures accessible		
Minimal capture of images/data		
De-identified public materials & approvals		

S4.3. Interview Protocols

Format: Semi-structured; 20–30 minutes; audio transcribed and anonymized.
Opening: State purpose, anonymity, and right to withdraw.

S4.3.1 Teacher Interviews (pre/post versions)

Pre (focus on current issues in misalignment/instability/missing evaluation)

1. Prior experience and challenges using digital/AI tools in labs? (misalignment & workload)
2. Common sources of instability? Usual responses? (devices/network/materials)
3. Understanding and practice regarding "evidence chains" and formative assessment?
4. Concerns and needs about safety and privacy?

Post (focus on mechanisms and improvement)

1. How do you understand this term's TPACK × SAMR alignment? Give examples of design changes.
2. How did dual-channel fallback (local/cloud; script/Excel) work during instability? Trade-offs and benefits?
3. After adding "uncertainty/error sources" to rubrics, how did student work change?
4. Which practices are most reusable? What are barriers to cross-class/grade adoption?
5. Suggestions regarding RAMP and privacy-compliance processes?

S4.3.2 Student Focus Groups (4–6 per group)

1. Which step most helped you understand *why* you did it? (purpose–method–evidence)

2. When tech issues occurred, how did your team keep moving? (fallback experience)
3. Difficulties and gains when adding “uncertainty/error sources” to the report?
4. Preferred ways to present results (oral/graphs/poster)? (UDL—multiple means of expression)
5. Your understanding and views on safety and privacy (e.g., open classes/public sharing)?

S4.3.3 Peer-Observer Short Interviews

1. How well aligned were lesson plans and enactment? Where and why did deviations occur?
2. Did formative tools genuinely change student interaction and product quality?
3. Did stability and fallback affect instructional pacing?
4. Which practices can you transfer to your own classes?

S4.4. Initial Qualitative Coding Frame (iterative)

Theme Code	Definition	Typical Evidence	Linked Challenge
T1_Alignment	Goal–task–assessment consistency; explicit TPACK	Alignment among objective cards, task sheets, rubric fields	Misalignment
T2_StabilityRecovery	Switching within 5 minutes after a failure; sustaining instruction	Switch logs; backup Excel/local scripts	Instability
T3_EvidenceChain	Data → analysis → conclusion is traceable; includes uncertainty	Report sections; peer-review records	Missing evaluation
T4_PublicOutput	Audience-facing presentation and revisions	Posters/defense; review comments and revised versions	Missing evaluation / Growth
T5_ComplianceSafety	RAMP execution; privacy de-identification and review	Audit sheets; consent forms; de-identified samples	Instability / Growth

Recommendation: Use **S4.4** as an open-coding guide first; after ≥6 lessons of field material, converge themes and develop an operational manual.

Use & Archiving Recommendations

- Collect rubrics and observation forms via **electronic forms**; export **CSV** for statistics and audit.
- Transcribe interviews, **de-identify** them, and archive the **codebook** and **audit trail** (rater disagreements, adjudications, versioning).
- Within the school’s shared repository, maintain versioned “**script–data–rubric–evidence**” bundles to support reuse and scale-up.